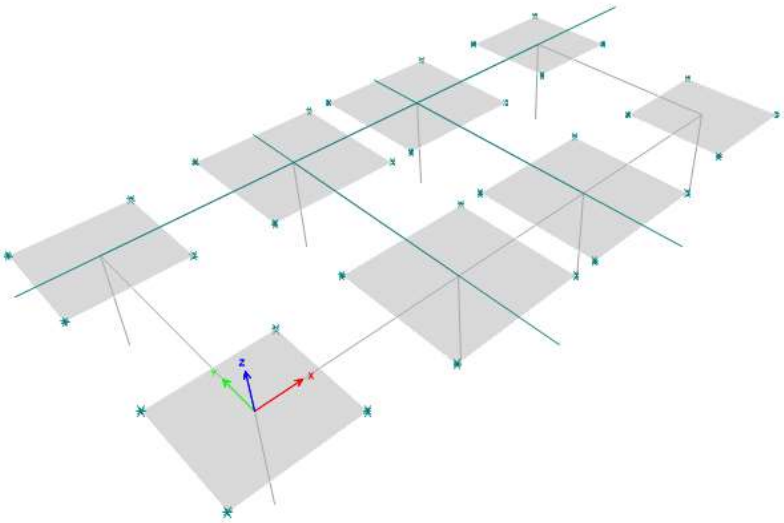

User Report 4

G+4_BUILDING

Model File: G+4_BUILDING_MODEL, Revision

06-03-2026

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1 Structure Data

This chapter provides model geometry information, including items such as story levels, point coordinates, and element connectivity.

1.1 Grid Data

Table 1.1 - Grid Definitions - General

Name	Type	Ux m	Uy m	Rz deg	Datum Name	Story Name Above	Story Name Below	Model Datum m	Story Height Above m	Story Height Below m	Bubble Size mm	Color
GLOBAL	Cartesian	0	0	0	Story1		Base	0		3	1250	10461087

Table 1.2 - Grid Definitions - Grid Lines

Name	Grid Line Type	ID	Ordinate m	Bubble Location	Visible
GLOBAL	X (Cartesian)	A	0	End	Yes
GLOBAL	X (Cartesian)	B	5.5	End	Yes
GLOBAL	X (Cartesian)	C	10	End	Yes
GLOBAL	X (Cartesian)	D	15.5	End	Yes
GLOBAL	Y (Cartesian)	1	0	End	Yes
GLOBAL	Y (Cartesian)	2	6.5	End	Yes

1.2 Area Connectivity

Table 1.3 - Floor Object Connectivity

Unique Name	UniquePt1	UniquePt2	UniquePt3	UniquePt4	Perimeter m	Area m2
1	1	2	3	4	12.4	9.61
3	9	10	11	12	12.4	9.61
9	33	34	35	36	15.2	14.44
10	37	38	39	40	15.2	14.44

1.3 Mass

Table 1.4 - Mass Source Definition

Name	Is Default	Include Lateral Mass?	Include Vertical Mass?	Lump Mass?	Source Self Mass?	Source Added Mass?	Source Load Patterns?	Move Mass Centroid?
MsSrc1	Yes	No	Yes	Yes	Yes	Yes	No	No

Table 1.5 - Mass Summary by Group

Group	Self Mass kg	Mass X kg	Mass Y kg	Mass Z kg
ALL	161821.95	161821.95	161821.95	161821.95

2 Properties

This chapter provides property information for materials, frame sections, shell sections, and links.

2.1 Materials

Table 2.1 - Material Properties - General

Material	Type	SymType	Grade	Color	Notes
A416MGr186	Tendon	Uniaxial	Unknown	Magenta	ASTM A416 Metric Grade 186 added 05-03-2026 18:34:50
CSA-G30.18Gr400	Rebar	Uniaxial	Unknown	Cyan	CSA-G30.18 Grade 400 added 05-03-2026 18:34:50
HYSD500	Rebar	Uniaxial	HYSD Grade 500	Blue	
M20	Concrete	Isotropic	M20	Gray8Dark	

2.2 Shell Sections

Table 2.2 - Area Section Property Definitions - Summary

Name	Type	Element Type	Material	Total Thickness mm
FOOTING600mm	Slab	Shell-Thin	M20	600
FOOTING700mm	Slab	Shell-Thin	M20	700

2.3 Reinforcement Sizes

Table 2.3 - Reinforcing Bar Sizes

Name	Diameter mm	Area cm2
10	10	0.8

2.4 Spring Properties

Table 2.4 - Spring Property Definitions - Area Springs

Name	Stiffness Option	Stiffnes U1 kN/m/m2	Stiffnes U2 kN/m/m2	Stiffnes U3 kN/m/m2	Nonlinear Option for U3	Color	Notes
SOIL1	User	0	0	8000	Tension Only	Cyan	

3 Assignments

This chapter provides a listing of the assignments applied to the model.

3.1 Joint Assignments

Table 3.1 - Joint Assignments - Summary

UniqueName	Diaphragm	Restraints	Include in Mesh
42	From Area		Yes
44	From Area		Yes
46	From Area		Yes
48	From Area		Yes
1	From Area	UX; UY	Yes
2	From Area	UX; UY	Yes
3	From Area	UX; UY	Yes
4	From Area	UX; UY	Yes
9	From Area	UX; UY	Yes
10	From Area	UX; UY	Yes
11	From Area	UX; UY	Yes
12	From Area	UX; UY	Yes
33	From Area	UX; UY	Yes
34	From Area	UX; UY	Yes
35	From Area	UX; UY	Yes
36	From Area	UX; UY	Yes
37	From Area	UX; UY	Yes
38	From Area	UX; UY	Yes
39	From Area	UX; UY	Yes
40	From Area	UX; UY	Yes
57	From Area		
58	From Area		
59	From Area		
60	From Area		

3.2 Shell Assignments

Table 3.2 - Area Assignments - Summary

UniqueName	Section Property	Property Type	Spring
1	FOOTING600mm	Slab	SOIL1
3	FOOTING600mm	Slab	SOIL1
9	FOOTING700mm	Slab	SOIL1
10	FOOTING700mm	Slab	SOIL1

4 Loads

This chapter provides loading information as applied to the model.

4.1 Load Patterns

Table 4.1 - Load Pattern Definitions

Name	Is Auto Load	Type	Self Weight Multiplier
~LLRF	No	Other	0
Dead	No	Dead	1
Dead $_{ABOVE}$	No	Dead	0
Live	No	Live	0
Live $_{ABOVE}$	No	Live	0

4.2 Load Cases

Table 4.2 - Load Case Definitions - Summary

Name	Type
Dead	Linear Static
Live	Linear Static

4.3 Load Combinations

Table 4.3 - Load Combination Definitions

Name	Type	Is Auto	Load Name	SF	Notes
DConS1	Linear Add	No	Dead	1.5	DConS1
DConS2	Linear Add	No	Dead	1.5	DConS2
DConS2			Live	1.5	

5 Analysis Results

This chapter provides analysis results.

5.1 Structure Results

Table 5.1 - Base Reactions

Output Case	Case Type	FX kN	FY kN	FZ kN	MX kN-m	MY kN-m	MZ kN-m	X m	Y m	Z m
Dead	LinStatic	-48	-48	11888.3113	39093.0116	-92686.4125	-264	0	0	-3
Live	LinStatic	0	0	1511.25	4911.5624	-11712.1874	0	0	0	-3
DConS1	Combination	-72	-72	17832.467	58639.5174	-139029.6187	-396	0	0	-3
DConS2	Combination	-72	-72	20099.3419	66006.861	-156597.8998	-396	0	0	-3

6 Design Data

This chapter provides design data and results.

6.1 Concrete Slab Design

Table 6.1 - Concrete Design Preferences - IS 456-2000

Item	Value
GammaSteel	1.15
GammaConc	1.5
IncreaseFlexRebar	No
Ignore Pu?	Yes
Pattern Live Load Factor	0.75
CoverTop mm	50
CoverBot mm	50
BarSize	10
InnerLayer	Layer B
PTCGSTop mm	25
PTCGSBotExt mm	40
PTCGSBotInt mm	25
SlabType	Two Way
Cover Beam Top mm	40
Cover Beam Bottom mm	40
Bar Size Beam Flexure	28
Bar Size Beam Shear	12
PTCGS Beam Top mm	50
PTCGS Beam Bottom mm	50
UserStress	No
InitConcRat	0.8
LLFraction	0.25

Table 6.2 - Concrete Slab Design Overwrites - Finite Element Based

Unique Name	Rebar Material	Cover Specification Type	LLRF	Design?	Ignote PT?
1	CSA-G30.18Gr400	From Preferences	1	Yes	No
3	CSA-G30.18Gr400	From Preferences	1	Yes	No
9	CSA-G30.18Gr400	From Preferences	1	Yes	No
10	CSA-G30.18Gr400	From Preferences	1	Yes	No

Table 6.3 - Concrete Slab Design Overwrites - Punching Shear - General

Unique Name	Check Punching Shear?	Location Type	Perimeter Type	Effective Depth Type	Opening Definition	Reinforcement Type
42	Program Determined	Auto	Auto	Auto	Auto	None
44	Program Determined	Auto	Auto	Auto	Auto	None
46	Program Determined	Auto	Auto	Auto	Auto	None
48	Program Determined	Auto	Auto	Auto	Auto	None